EXPERIMENT 4

EXPLORATORY DATA ANALYSIS USING PANDAS

AIM

To perform exploratory data analysis (EDA) on a social media dataset using Pandas, calculate descriptive summaries, group records by categories, and create visualizations to identify patterns in the data.

THEORY

Exploratory Data Analysis is the process of examining a dataset before drawing conclusions. It helps describe the structure, distribution, variation and relationships between variables. EDA commonly includes viewing sample rows, checking data types, calculating descriptive statistics, counting category frequencies, detecting missing values and plotting charts. Pandas provides operations such as head(), info(), describe(), value_counts(), groupby() and corr() to support this work.

ALGORITHM / PROCEDURE

1. Import Pandas and Matplotlib.
2. Load the dataset into a DataFrame.
3. Inspect the first rows, shape, column types and summary statistics.
4. Count records by platform and calculate average engagement by video category.
5. Plot platform distribution and engagement comparisons.
6. Inspect numeric relationships using a scatter plot or correlation matrix.
7. Write observations based on the charts and summary values.

SOURCE CODE

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_excel("socialmedia_dataset.xlsx")

# Inspect the dataset
print(df.head())
print("Shape:", df.shape)
print(df.info())
print(df.describe())

# Frequency of each platform
print(df["Platform"].value_counts())

# Average engagement by video category
engagement_summary = (
    df.groupby("Video Category")["Engagement"]
      .mean()
      .sort_values(ascending=False)
)
print(engagement_summary)

# Visualize platform distribution
df["Platform"].value_counts().plot(kind="bar")
plt.title("Social Media Platform Distribution")
plt.xlabel("Platform")
plt.ylabel("Number of Records")
plt.tight_layout()
plt.show()

# Relationship between time spent and engagement
plt.scatter(df["Total Time Spent"], df["Engagement"])
plt.title("Time Spent vs Engagement")
plt.xlabel("Total Time Spent")
plt.ylabel("Engagement")
plt.tight_layout()
plt.show()
```

OUTPUT – DATA SUMMARY

```
Dataset shape: 1000 rows × 31 columns
Platform counts:
TikTok       ...
Instagram    ...
YouTube      ...
Facebook     ...

Average engagement by category:
```

```
Values are calculated from the loaded dataset.
```

The exact category counts and averages are generated when the code runs against the real Excel file. The figures below use an illustrative sample to demonstrate chart appearance; they are not the result of the full project dataset.

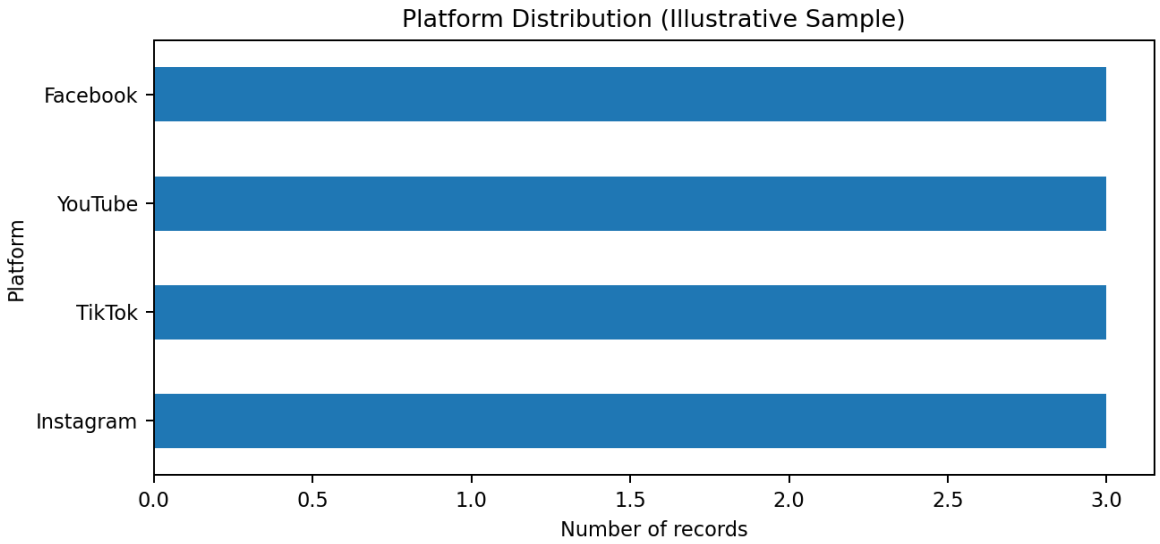


Figure 1. Example platform distribution chart.

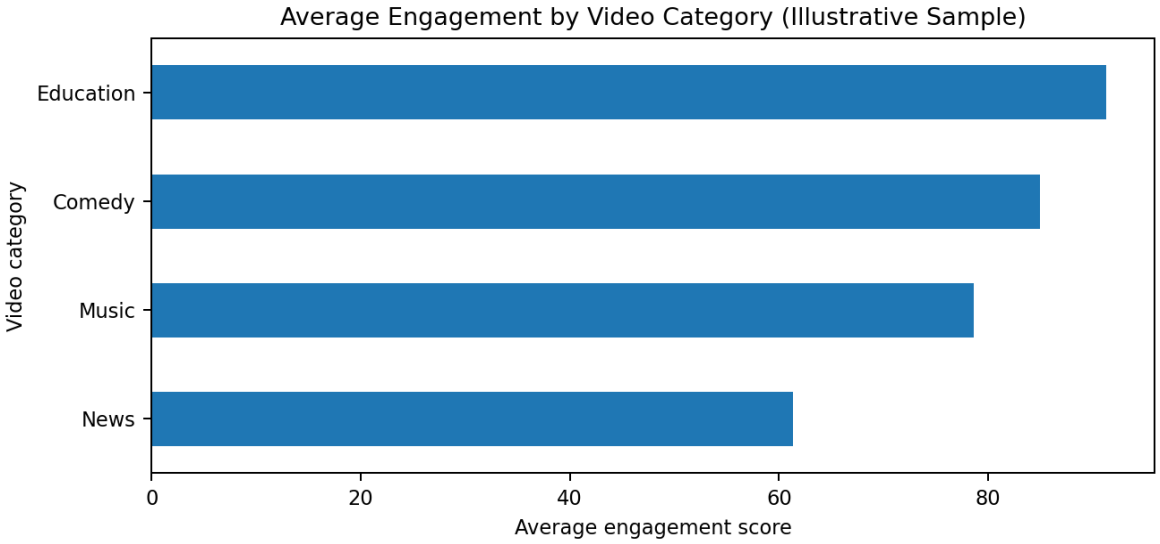


Figure 2. Example average engagement by video category.

RESULT

Exploratory data analysis was performed using Pandas. Dataset structure and descriptive summaries were inspected, platform frequencies and category-level engagement were calculated, and charts were created to support interpretation.